MATHEMATICS / INDEPENDENT STUDY

# MM8036

# Type Theory

Progressive hints

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Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

Edition 2 · 25 October 2026

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**How to use this volume** E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give an argument and criteria for self-assessment. Expanded solutions also explain how to find the approach, a tempting dead end, and the role of the hypotheses.

Each unit has twelve core problems (four E, five C, three A) and an optional A-level transfer challenge numbered 13. The challenge asks you to choose a method in a new setting; budget its hour separately. Problem 12 and the challenge have detailed solutions with independently written decisive ideas. Mock estimates total 170 minutes of problem work plus ten minutes for review.

# 1 Unit 1 · Binding and substitution

### MM8036-U1.01 · E · 15 min
Learning objective: U1.1 Parsing

Fully parenthesize $\lambda x.xyz$ and $ab(cd)$.

**Hint 1 — a starting point.** Application associates left.

**Hint 2 — the key move.** The abstraction body extends to the right.

### MM8036-U1.02 · E · 15 min
Learning objective: U1.1 Free variables

Compute $\mathrm{FV}(\lambda x.(xy)(\lambda y.yz))$.

**Hint 1 — a starting point.** Compute each subtree first.

**Hint 2 — the key move.** The inner binder removes only its own $y$.

### MM8036-U1.03 · E · 15 min
Learning objective: U1.1 Alpha equivalence

Which of $\lambda x.\lambda y.x$, $\lambda a.\lambda b.a$, and $\lambda a.\lambda b.b$ are alpha equivalent?

**Hint 1 — a starting point.** Track binder positions rather than names.

**Hint 2 — the key move.** Does the body refer to the outer or inner binder?

### MM8036-U1.04 · E · 15 min
Learning objective: U1.2 Substitution

Compute $(\lambda y.xy)[x := yz]$.

**Hint 1 — a starting point.** The binder conflicts with a free variable of the argument.

**Hint 2 — the key move.** Rename it to fresh $w$.

### MM8036-U1.05 · C · 30 min
Learning objective: U1.2 Substitution under shadowing

Compute $(\lambda x.(\lambda y.xy)z)[y := x]$ and explain why the replacement has no effect.

**Hint 1 — a starting point.** Find the free occurrences of $y$.

**Hint 2 — the key move.** Every $y$ in this term is bound.

### MM8036-U1.06 · C · 30 min
Learning objective: U1.2 Exact free-variable formula

Prove: if $x \in \mathrm{FV}(t)$, then $\mathrm{FV}(t[x := u]) = (\mathrm{FV}(t) \setminus \{x\}) \cup \mathrm{FV}(u)$.

**Hint 1 — a starting point.** Strengthen the usual inclusion by considering whether the variable occurs.

**Hint 2 — the key move.** Induct on syntax; in the application case at least one branch contains a free $x$.

### MM8036-U1.07 · C · 30 min
Learning objective: U1.2 Closed substitutions

If $t$ has free variables among $x_1, \ldots, x_n$ and each $u_i$ is closed, prove simultaneous substitution of $u_i$ for $x_i$ closes $t$.

**Hint 1 — a starting point.** Use the free-variable bound.

**Hint 2 — the key move.** Closed replacements cannot introduce new free variables.

### MM8036-U1.08 · C · 30 min
Learning objective: U1.2 Composition law

Verify the substitution-composition law for $t = xy$, $u = yz$, $v = w$, substituting first $x$ then $y$.

**Hint 1 — a starting point.** Check $x \neq y$ and $x \notin \mathrm{FV}(w)$.

**Hint 2 — the key move.** Evaluate both sides separately.

### MM8036-U1.09 · C · 30 min
Learning objective: U1.1 Structural induction

Let size count one for each variable, application, and abstraction node. Prove $|t[x := u]| = |t| + k(|u| - 1)$, where $k$ is the number of free occurrences of $x$ in $t$.

**Hint 1 — a starting point.** Count what happens at each replaced variable leaf.

**Hint 2 — the key move.** Renaming changes names but not node counts.

### MM8036-U1.10 · A · 50 min
Learning objective: U1.2 Necessity of hypotheses

Construct a counterexample to the substitution-composition equation when $x \in \mathrm{FV}(v)$, even though $x \neq y$.

**Hint 1 — a starting point.** Use $t = y$.

**Hint 2 — the key move.** Take $u = z$ and $v = x$ with three distinct names.

### MM8036-U1.11 · A · 50 min
Learning objective: U1.2 Commuting substitutions

Prove that substitutions for distinct $x, y$ commute if $x \notin \mathrm{FV}(v)$ and $y \notin \mathrm{FV}(u)$: $(t[x := u])[y := v] = (t[y := v])[x := u]$ up to alpha equivalence. Explain why both side conditions are needed.

**Hint 1 — a starting point.** Apply composition and simplify its replacement term.

**Hint 2 — the key move.** Use the unchanged-term lemma on $u[y := v]$.

### MM8036-U1.12 · A · 50 min
Learning objective: U1.1 Binding invariants

Use zero-based de Bruijn indices to encode the closed terms $\lambda x.\lambda y.x(yx)$ and $\lambda a.\lambda b.a(ba)$. Explain why this representation detects their alpha equivalence, and why a rule for free variables is still needed for open terms.

**Hint 1 — a starting point.** Index zero denotes the nearest enclosing binder.

**Hint 2 — the key move.** The body is $1(01)$.

### MM8036-U1.13 · A · 60 min · Optional challenge

Learning objective: U1 Transfer: Two ways to replace names

For $t = \lambda z.xz$, compute first $t[x := y][y := z]$ and then the single simultaneous substitution $t[x := y, y := z]$. Are the results alpha equivalent? Explain every binder renaming.

**Hint 1 — a starting point.** Sequential substitution also acts on the result of the preceding substitution.

**Hint 2 — the key move.** A simultaneous substitution inserts each specified image once; it does not recursively substitute inside that image.

# 2 Unit 2 · Reduction and confluence

### MM8036-U2.01 · E · 15 min
Learning objective: U2.1 Beta reduction

Reduce $(\lambda x.x)(\lambda y.y)$ to normal form.
**Hint 1 — a starting point.** Locate the outer redex.
**Hint 2 — the key move.** Substitute the argument for $x$.

### MM8036-U2.02 · E · 15 min
Learning objective: U2.1 Capture avoidance

Reduce $(\lambda x.\lambda y.x)y$.
**Hint 1 — a starting point.** The free argument must remain free.
**Hint 2 — the key move.** Rename the inner binder before contraction.

### MM8036-U2.03 · E · 15 min
Learning objective: U2.1 Divergence

Let $\Delta = \lambda x.xx$ and $\Omega = \Delta\Delta$. Show an infinite beta sequence from $\Omega$.
**Hint 1 — a starting point.** Contract its outer redex.
**Hint 2 — the key move.** The contractum is identical to the source.

### MM8036-U2.04 · E · 15 min
Learning objective: U2.1 Reduction strategies

Compare outermost reduction and call-by-value on $(\lambda x.\lambda z.z)\Omega$.
**Hint 1 — a starting point.** The function ignores its argument.
**Hint 2 — the key move.** Call-by-value evaluates the argument first.

### MM8036-U2.05 · C · 30 min
Learning objective: U2.2 Joining a fork

For $(\lambda x.xx)((\lambda y.y)z)$, contract either the outer redex or the argument redex first and exhibit a common reduct.
**Hint 1 — a starting point.** The outer step duplicates the argument.
**Hint 2 — the key move.** Reduce both duplicated argument redexes.

### MM8036-U2.06 · C · 30 min
Learning objective: U2.2 Normal-form uniqueness

Derive uniqueness of beta normal forms up to alpha equivalence from confluence.
**Hint 1 — a starting point.** Apply confluence to two normal reducts of one term.
**Hint 2 — the key move.** A normal term cannot take a positive number of beta steps.

### MM8036-U2.07 · C · 30 min
Learning objective: U2.2 Conversion

Show that beta-convertible normal terms are alpha equivalent. You may use Church–Rosser for conversion.

**Hint 1 — a starting point.** Conversion can contain reversed steps.

**Hint 2 — the key move.** Church–Rosser replaces the zigzag by a common reduct.

### MM8036-U2.08 · C · 30 min
Learning objective: U2.1 Weak versus strong normalization

Prove that $(\lambda x.\lambda z.z)\Omega$ is weakly normalizing but not strongly normalizing under full beta reduction.

**Hint 1 — a starting point.** Use two different permitted sequences.

**Hint 2 — the key move.** Full reduction may reduce inside the ignored argument.

### MM8036-U2.09 · C · 30 min
Learning objective: U2.2 Parallel reduction

Compute the complete development of $(\lambda x.xx)((\lambda y.y)z)$. Explain why it differs from contracting just one redex.

**Hint 1 — a starting point.** Develop the body and argument before substituting.

**Hint 2 — the key move.** The argument develops to $z$.

### MM8036-U2.10 · A · 50 min
Learning objective: U2.2 Confluence versus termination

Give a finite abstract reduction system that is confluent but has an infinite reduction sequence, and prove both claims.

**Hint 1 — a starting point.** A self-loop is enough, but add a visible normal form.

**Hint 2 — the key move.** Use $a \to a$ and $a \to b$ with $b$ normal.

### MM8036-U2.11 · A · 50 min
Learning objective: U2.2 Substitution and reduction

Prove that if $u \to u'$, then $t[x := u] \xrightarrow{*} t[x := u']$. Explain why one step is generally insufficient.

**Hint 1 — a starting point.** Induct on $t$, renaming binders fresh.

**Hint 2 — the key move.** There is one copy of the step for each free occurrence of $x$.

### MM8036-U2.12 · A · 50 min
Learning objective: U2.2 Distinguishing convertibility

Prove the closed terms $\lambda x.\lambda y.x$ and $\lambda x.\lambda y.y$ are not beta convertible. Why is their different behavior on some arguments not needed for this proof?

**Hint 1 — a starting point.** Inspect normal forms and binding structure.

**Hint 2 — the key move.** Combine normal-form uniqueness for conversion with the de Bruijn invariant.

### MM8036-U2.13 · A · 60 min · Optional challenge

Learning objective: U2 Transfer: Termination after duplicating an input

Let $I = \lambda z.z$ and $D = \lambda x.xx$. Suppose a closed term t reduces in finitely many beta steps to I. Prove that D t has a normal form. Must D t be strongly normalizing for every such t?

**Hint 1 — a starting point.** First follow the given finite reduction inside the argument of D.

**Hint 2 — the key move.** For the second part, an argument can reach I while containing a subterm whose reduction never stops.

# 3 Unit 3 · Simple types and judgments

### MM8036-U3.01 · E · 15 min
Learning objective: U3.1 Typing judgments
Give a derivation of $\lambda x.x : A \to A$ in the empty context.
**Hint 1 — a starting point.** Start with the variable rule.
**Hint 2 — the key move.** Discharge $x : A$ by abstraction.

### MM8036-U3.02 · E · 15 min
Learning objective: U3.1 Application
In context $f : A \to B, x : A$, derive $fx : B$.
**Hint 1 — a starting point.** Use the variable rule twice.
**Hint 2 — the key move.** Apply the arrow elimination rule.

### MM8036-U3.03 · E · 15 min
Learning objective: U3.1 Arrow association
Give a type and derivation for $\lambda x.\lambda y.x$.
**Hint 1 — a starting point.** Arrows associate right.
**Hint 2 — the key move.** First abstract over $y$, then $x$.

### MM8036-U3.04 · E · 15 min
Learning objective: U3.2 Substitution
Given $x : A \vdash x : A$ and $\vdash u : A$, calculate the conclusion of substitution.
**Hint 1 — a starting point.** Replace the free variable.
**Hint 2 — the key move.** The result type remains $A$ in simple typing.

### MM8036-U3.05 · C · 30 min
Learning objective: U3.1 Composition
Derive the type of $\lambda f.\lambda g.\lambda x.f(gx)$ from chosen types $f : B \to C, g : A \to B$.
**Hint 1 — a starting point.** Work from the innermost application outward.
**Hint 2 — the key move.** Discharge assumptions in reverse order.

### MM8036-U3.06 · C · 30 min
Learning objective: U3.2 Weakening
Prove admissibility of adding a fresh unused variable $y : B$ to a simply typed context.
**Hint 1 — a starting point.** Induct on the typing derivation.
**Hint 2 — the key move.** Rename abstraction binders to remain fresh for the new context.

### MM8036-U3.07 · C · 30 min
Learning objective: U3.2 Preservation

Prove preservation for a root redex $(\lambda x.t)u \to t[x := u]$ using inversion and substitution.
**Hint 1 — a starting point.** Invert application, then abstraction.
**Hint 2 — the key move.** The body premise has the exact assumption required by substitution.

### MM8036-U3.08 · C · 30 min
Learning objective: U3.2 Open terms and progress

Give a well-typed open term that is neither a call-by-value value nor able to step, under the lecture convention that values are abstractions and the inert base constant.
**Hint 1 — a starting point.** A neutral application has no visible function constructor.
**Hint 2 — the key move.** Use $f : o \to o, x : o$.

### MM8036-U3.09 · C · 30 min
Learning objective: U3.1 Untypability of self-application

Prove $\lambda x.xx$ is untypable in finite simple types.
**Hint 1 — a starting point.** Invert the two uses of $x$ in one context.
**Hint 2 — the key move.** Its type would have to equal an arrow having itself as domain.

### MM8036-U3.10 · A · 50 min
Learning objective: U3.2 Failure of subject expansion

Construct a beta step from an untypable term to a typable closed term. Prove both typing assertions.
**Hint 1 — a starting point.** An ignored argument may itself be untypable.
**Hint 2 — the key move.** Use $(\lambda x.\lambda y.y)\Omega$.

### MM8036-U3.11 · A · 50 min
Learning objective: U3.2 Substitution proof

Prove the application and abstraction cases of the substitution lemma, including why a binder may need renaming.
**Hint 1 — a starting point.** State the induction hypothesis for each immediate premise.
**Hint 2 — the key move.** For an abstraction, choose its binder fresh for the substituting term and distinct from the variable being removed.

### MM8036-U3.12 · A · 50 min
Learning objective: U3.1 Typing restricts syntax

Prove that a closed beta-normal arrow-only term of base type $o$, in the language with exactly one inert constant $c : o$, must be $c$.
**Hint 1 — a starting point.** Classify normal terms into abstractions and neutral spines.
**Hint 2 — the key move.** A closed neutral spine must be headed by a constant.

### MM8036-U3.13 · A · 60 min · Optional challenge

Learning objective: U3 Transfer: All normal outputs in a tiny context

In the arrow-only core with sole constant $c : o$, classify all beta-normal terms of type o in context $f : o \to o, x : o$. Prove your list is exhaustive and that distinct listed terms are not beta convertible.

**Hint 1 — a starting point.** Inspect the head of a normal application spine.

**Hint 2 — the key move.** The head can be x, c, or f; only f accepts an argument, again of type o.

# 4 Unit 4 · Normalization and constructive logic

**MM8036-U4.01 · E · 15 min**
Learning objective: U4.2 Curry–Howard
Write a proof term for $A \to B \to A$.
**Hint 1 — a starting point.** Introduce both assumptions.
**Hint 2 — the key move.** Return the first assumption.

**MM8036-U4.02 · E · 15 min**
Learning objective: U4.2 Conjunction
Construct $(A \times B) \to A$ and $(A \times B) \to B$.
**Hint 1 — a starting point.** Use product elimination.
**Hint 2 — the key move.** Use the two projections.

**MM8036-U4.03 · E · 15 min**
Learning objective: U4.2 Disjunction
Construct $A \to A + B$ and explain what data the output contains.
**Hint 1 — a starting point.** Choose the left constructor.
**Hint 2 — the key move.** Keep the supplied evidence for $A$.

**MM8036-U4.04 · E · 15 min**
Learning objective: U4.1 Normalization vocabulary
State weak and strong normalization and their implication relation.
**Hint 1 — a starting point.** One asks for a successful path; the other quantifies over all paths.
**Hint 2 — the key move.** Every infinite path must be excluded for strong normalization.

**MM8036-U4.05 · C · 30 min**
Learning objective: U4.2 Negation
Construct $(A \to B) \to ((B \to 0) \to (A \to 0))$.
**Hint 1 — a starting point.** Assume evidence for $A$ and aim at the empty type.
**Hint 2 — the key move.** Compose the two functions.

**MM8036-U4.06 · C · 30 min**
Learning objective: U4.2 Sum elimination
Construct $((A \to C) \times (B \to C)) \to ((A + B) \to C)$ and state its two computation cases.
**Hint 1 — a starting point.** Project the two functions and inspect the sum.
**Hint 2 — the key move.** Apply the function matching the tag.

### MM8036-U4.07 · C · 30 min
Learning objective: U4.1 Candidate closure

Explain why every variable belongs to every reducibility candidate defined in Lecture 7.
**Hint 1 — a starting point.** Use neutral closure, not a typing assignment.
**Hint 2 — the key move.** A variable has no immediate reducts.

### MM8036-U4.08 · C · 30 min
Learning objective: U4.1 Why strengthen induction

Show that strong normalization of $s$ and $t$ does not imply strong normalization of $st$.
**Hint 1 — a starting point.** Use a normal abstraction that duplicates its argument.
**Hint 2 — the key move.** Take both terms to be $\Delta = \lambda x.xx$.

### MM8036-U4.09 · C · 30 min
Learning objective: U4.2 Double negation

Construct $A \to ((A \to 0) \to 0)$ and explain why reversing the arrow requires a different argument.
**Hint 1 — a starting point.** An assumption of $A$ can be fed to its negation.
**Hint 2 — the key move.** Return $\lambda n.na$.

### MM8036-U4.10 · A · 50 min
Learning objective: U4.1 Canonicity consequence

Using normalization, preservation, and U3.12, prove every closed term of type $o$ reduces to $c$ in the arrow-only calculus with sole constant $c : o$.
**Hint 1 — a starting point.** Each theorem supplies one step in the argument.
**Hint 2 — the key move.** Normalize first, preserve the type, then classify the normal form.

### MM8036-U4.11 · A · 50 min
Learning objective: U4.2 Constructive equivalence

Construct maps in both directions between $(A + B) \to C$ and $(A \to C) \times (B \to C)$. Show the composites agree pointwise on constructor inputs.
**Hint 1 — a starting point.** Restrict a function along each injection; reverse by cases.
**Hint 2 — the key move.** Test the resulting functions at left and right inputs.

### MM8036-U4.12 · A · 50 min
Learning objective: U4.1 Proof reconstruction: normalization dependencies

Reconstruct the abstraction case of the fundamental reducibility lemma. Identify exactly where the controlled beta-expansion lemma is used.
**Hint 1 — a starting point.** The arrow clause needs strong normalization of the abstraction and reducibility of every application.
**Hint 2 — the key move.** Use two environment extensions: a fresh variable and an arbitrary reducible argument.

### MM8036-U4.13 · A · 60 min · Optional challenge

Learning objective: U4 Transfer: Composing reducible raw terms

Let $t \in \mathcal{R}_{A\to B}$ and $u \in \mathcal{R}_{B\to C}$, with x fresh for both. Prove that $\lambda x.u(tx)$ belongs to $\mathcal{R}_{A\to C}$. Do not assume t or u has a typing derivation.

**Hint 1 — a starting point.** A fresh variable belongs to every candidate.

**Hint 2 — the key move.** First prove the body strongly normalizing; then test the abstraction on an arbitrary reducible argument.

# 5 Unit 5 · Dependent types

### MM8036-U5.01 · E · 15 min
Learning objective: U5.1 Pi formation

State the data needed to form $\Pi_{x:A} B(x)$ in context $\Gamma$.
**Hint 1 — a starting point.** The codomain is checked in an extended context.
**Hint 2 — the key move.** It may mention the new variable $x$.

### MM8036-U5.02 · E · 15 min
Learning objective: U5.1 Dependent application

If $f : \Pi_{x:A} B(x)$ and $a : A$, what is the type of $fa$?
**Hint 1 — a starting point.** Substitute the actual argument into the family.
**Hint 2 — the key move.** The result lies over $a$.

### MM8036-U5.03 · E · 15 min
Learning objective: U5.2 Sigma introduction

What must be supplied to construct a term of $\Sigma_{x:A} B(x)$?
**Hint 1 — a starting point.** The second component depends on the first.
**Hint 2 — the key move.** Supply $a : A$ and then $b : B(a)$.

### MM8036-U5.04 · E · 15 min
Learning objective: U5.2 Sigma projections

State the types and computation rules for the projections of $p : \Sigma_{x:A} B(x)$.
**Hint 1 — a starting point.** The second projection's type mentions the first.
**Hint 2 — the key move.** Evaluate both projections on $(a, b)$.

### MM8036-U5.05 · C · 30 min
Learning objective: U5.1 Dependent identity

Derive the type of $\lambda A.\lambda x.x$ with $A : U_0$. Determine the level of its type.
**Hint 1 — a starting point.** The inner identity is small, but the outer domain is a universe.
**Hint 2 — the key move.** Use $U_0 : U_1$.

### MM8036-U5.06 · C · 30 min
Learning objective: U5.2 Dependent pair map

Given $f : A \to A'$ and $g : \Pi_{x:A}(B(x) \to B'(fx))$, construct a map $\Sigma_{x:A} B(x) \to \Sigma_{y:A'} B'(y)$.
**Hint 1 — a starting point.** Eliminate the input pair.
**Hint 2 — the key move.** Send $(a, b)$ to $(fa, gab)$.

**MM8036-U5.07 · C · 30 min**
Learning objective: U5.1 Dependent composition
Given $f : \Pi_{x:A} B(x)$ and $g : \Pi_{x:A}(B(x) \to C(x))$, derive a term of $\Pi_{x:A} C(x)$.
**Hint 1 — a starting point.** Work under an arbitrary $x : A$.
**Hint 2 — the key move.** Apply $g$ at the same index as $f$.

**MM8036-U5.08 · C · 30 min**
Learning objective: U5.2 Existential evidence
Explain the computational difference between $\Sigma_{n:\mathrm{Nat}} P(n)$ and $\Pi_{n:\mathrm{Nat}} P(n)$, and give their respective elimination uses.
**Hint 1 — a starting point.** One stores a witness; one accepts an arbitrary input.
**Hint 2 — the key move.** Project a Sigma; apply a Pi.

**MM8036-U5.09 · C · 30 min**
Learning objective: U5.2 Constant families
Show that Sigma and Pi over constant families reproduce ordinary products and arrows.
**Hint 1 — a starting point.** Write their introduction, elimination, and computation rules.
**Hint 2 — the key move.** The substitution into a constant family changes nothing.

**MM8036-U5.10 · A · 50 min**
Learning objective: U5.2 Constructive choice
Construct a function from $\Pi_{x:A} \Sigma_{y:B(x)} R(x, y)$ to $\Sigma_{f:\Pi_{x:A} B(x)} \Pi_{x:A} R(x, fx)$. Explain why this does not introduce classical choice.
**Hint 1 — a starting point.** Each input already provides a pair at every $x$.
**Hint 2 — the key move.** Define $f$ by first projection and its evidence by second projection.

**MM8036-U5.11 · A · 50 min**
Learning objective: U5.2 Dependent currying
Construct maps between $\Pi_{p:\Sigma_{x:A} B(x)} C(\pi_1(p), \pi_2(p))$ and $\Pi_{x:A} \Pi_{y:B(x)} C(x, y)$. Verify their computation on pair inputs.
**Hint 1 — a starting point.** Evaluate an uncurried function on a constructed pair.
**Hint 2 — the key move.** For the reverse, eliminate the input pair.

**MM8036-U5.12 · A · 50 min**
Learning objective: U5.1 Context dependency
Explain why exchanging $x : A, y : B(x)$ into $y : B(x), x : A$ is generally invalid, although weakening by a fresh declaration is often admissible. Give a concrete family example.
**Hint 1 — a starting point.** A context must be well formed from left to right.
**Hint 2 — the key move.** Use $n : \mathrm{Nat}, v : \mathrm{Vec}(D, n)$.

### MM8036-U5.13 · A · 60 min · Optional challenge

Learning objective: U5 Transfer: Repackaging dependent records

For types $A$, $B(x)$, and $C(x,y)$, construct maps between $S = \Sigma_{x:A}\Sigma_{y:B(x)}C(x,y)$ and $T = \Sigma_{p:\Sigma_{x:A}B(x)}C(\mathrm{fst}(p),\mathrm{snd}(p))$. Check all projection types and compute both composites on constructor inputs.

**Hint 1 — a starting point.** Both records store the same three pieces of data, grouped differently.

**Hint 2 — the key move.** Track the third component's type after each projection; do not replace it by a constant type.

# 6 Unit 6 · Induction, identity, and universes

**MM8036-U6.01 · E · 15 min**
Learning objective: U6.1 Addition computation
With addition recursive in its first argument, compute $\mathrm{add}(\mathrm{suc}(\mathrm{zero}), n)$.
**Hint 1 — a starting point.** Use successor, then zero computation.
**Hint 2 — the key move.** Keep the variable $n$ untouched.

**MM8036-U6.02 · E · 15 min**
Learning objective: U6.1 Induction motive
State the motive for proving right-zero addition.
**Hint 1 — a starting point.** The statement must vary with the induction input.
**Hint 2 — the key move.** Use an identity type indexed by $n$.

**MM8036-U6.03 · E · 15 min**
Learning objective: U6.2 Transport typing
For $p : \mathrm{Id}_{A(a,b)}$ and $u : B(a)$, state the type of $\mathrm{transport}_{B(p,u)}$ and its reflexivity computation.
**Hint 1 — a starting point.** It moves evidence to the endpoint fiber.
**Hint 2 — the key move.** At reflexivity the endpoints coincide.

**MM8036-U6.04 · E · 15 min**
Learning objective: U6.2 Universe levels
If $A : U_1$ and $B(x) : U_0$, find a universe containing $\Pi_{x:A} B(x)$ and one containing $\mathrm{Id}_{A(a,b)}$.
**Hint 1 — a starting point.** Use maximum for Pi and the base level for identity.
**Hint 2 — the key move.** Both depend on the level of $A$.

**MM8036-U6.05 · C · 30 min**
Learning objective: U6.1 Associativity of addition
Prove addition associative by induction on its first argument.
**Hint 1 — a starting point.** Write the motive with the other two arguments fixed.
**Hint 2 — the key move.** Lift the induction equality through successor.

**MM8036-U6.06 · C · 30 min**
Learning objective: U6.1 Vector append
Describe and type both clauses of append on length-indexed vectors.
**Hint 1 — a starting point.** Recurse on the first vector, matching the recursion of addition.
**Hint 2 — the key move.** The nil case length converts from $0 + n$ to $n$.

### MM8036-U6.07 · C · 30 min
Learning objective: U6.2 Symmetry by J

Derive symmetry of identity with an explicit motive and reflexive case.
**Hint 1 — a starting point.** The desired output reverses endpoints.
**Hint 2 — the key move.** Set $C(a, b, p) = \mathrm{Id}_{A(b,a)}$.

### MM8036-U6.08 · C · 30 min
Learning objective: U6.2 Congruence

For $f : A \to D$, derive $\mathrm{Id}_{A(a,b)} \to \mathrm{Id}_{D(f(a),f(b))}$. Why is this an implication about identity proofs rather than an equality-reflection rule?
**Hint 1 — a starting point.** Use path induction on the input proof.
**Hint 2 — the key move.** The reflexive case is reflexivity at $f(a)$.

### MM8036-U6.09 · C · 30 min
Learning objective: U6.1 Safe head

Explain why a head function $\mathrm{Vec}(A, \mathrm{suc}(n)) \to A$ needs no nil branch. Can the same total signature be given for all lengths without extra data?
**Hint 1 — a starting point.** Constructor indices constrain which inputs can inhabit the domain.
**Hint 2 — the key move.** Nil has length zero.

### MM8036-U6.10 · A · 50 min
Learning objective: U6.2 Transport composition

For $p : \mathrm{Id}_{A(a,b)}$ and $q : \mathrm{Id}_{A(b,c)}$, prove transport along their composite agrees propositionally with successive transport on $u : B(a)$.
**Hint 1 — a starting point.** Induct on the paths, generalizing their endpoints appropriately.
**Hint 2 — the key move.** After both become reflexivity, both expressions compute to $u$.

### MM8036-U6.11 · A · 50 min
Learning objective: U6.2 Judgmental and propositional equality

Compare the left-zero and right-zero laws for addition. Explain why a proof of right-zero does not make its open expression reduce by definition.
**Hint 1 — a starting point.** Inspect which argument the recursive definition matches.
**Hint 2 — the key move.** A variable in the first argument blocks computation.

### MM8036-U6.12 · A · 50 min
Learning objective: U6.2 Diagnosing misuse of path induction

A student argues: "Apply J to $p : \mathrm{Id}_{A(a,a)}$, so every such $p$ equals reflexivity." Identify the missing motive issue; explain what J actually supplies.
**Hint 1 — a starting point.** J varies an endpoint before specializing to loops.
**Hint 2 — the key move.** The alleged comparison to reflexivity is not well typed when the endpoint varies.

### MM8036-U6.13 · A · 60 min · Optional challenge

Learning objective: U6 Transfer: A section respects changes of fiber

Let $B : A \to U$ and $f : \Pi_{x:A} B(x)$. Given $p : \mathrm{Id}_{A(a,b)}$, construct a term of $\mathrm{Id}_{B(b)}\left(\mathrm{transport}_{B(p)}(f(a)), f(b)\right)$ and state its computation when $p$ is reflexivity. Explain why simply comparing $f(a)$ with $f(b)$ would not generally give a well-formed identity type.

**Hint 1 — a starting point.** Check the type of each endpoint of the proposed identity before trying to supply its proof.

**Hint 2 — the key move.** Let the endpoint b and its equality p vary; when p is reflexivity, the transport disappears.

# 7 Unit 7 · Substitutions and semantics

**MM8036-U7.01 · E · 15 min**
Learning objective: U7.1 Context substitution
Describe a substitution from $\Delta$ to the context $x : A, y : B(x)$.
**Hint 1 — a starting point.** Supply terms in declaration order.
**Hint 2 — the key move.** The second term is checked after the first substitution.

**MM8036-U7.02 · E · 15 min**
Learning objective: U7.1 Identity substitution
Write the identity substitution on $x : A, y : B(x)$.
**Hint 1 — a starting point.** Use the context's own variables.
**Hint 2 — the key move.** Check the dependent second component.

**MM8036-U7.03 · E · 15 min**
Learning objective: U7.1 Composition order
For $\sigma : \Delta \to \Gamma$ and $\tau : \Theta \to \Delta$, state the source and target of $\sigma \circ \tau$ and the substitution equation for a term $t$ over $\Gamma$.
**Hint 1 — a starting point.** Read composition right to left.
**Hint 2 — the key move.** First reinterpret $t$ over $\Delta$, then over $\Theta$.

**MM8036-U7.04 · E · 15 min**
Learning objective: U7.2 Meaning explanations
Contrast a canonical natural-number term with a neutral natural-number term.
**Hint 1 — a starting point.** Canonical forms expose a constructor.
**Hint 2 — the key move.** A variable need not compute to a visible numeral in an open context.

**MM8036-U7.05 · C · 30 min**
Learning objective: U7.1 Associativity on components
Prove associativity of context-substitution composition from the term substitution law.
**Hint 1 — a starting point.** Compare the component corresponding to each declaration.
**Hint 2 — the key move.** Both composites substitute twice into the same supplied term.

**MM8036-U7.06 · C · 30 min**
Learning objective: U7.1 Projection and pairing
For context extension $\Gamma \cdot A$, verify $p \circ \langle \sigma, t \rangle = \sigma$ and $q[\langle \sigma, t \rangle] = t$ using tuple substitutions.
**Hint 1 — a starting point.** Pairing appends a last component.
**Hint 2 — the key move.** Projection discards that last component; the generic variable selects it.

### MM8036-U7.07 · C · 30 min
Learning objective: U7.2 Families of sets

In the set interpretation, describe the meaning of a dependent sum for $A = \{0, 1\}$, $B(0) = \{a\}$, $B(1) = \{b, c\}$. List its elements.

**Hint 1 — a starting point.** Keep tags identifying the starting fiber.

**Hint 2 — the key move.** It is a disjoint union of fibers.

### MM8036-U7.08 · C · 30 min
Learning objective: U7.2 Dependent products of sets

For the same two fibers as U7.07, list all dependent functions and count them.

**Hint 1 — a starting point.** Choose one value from each fiber.

**Hint 2 — the key move.** The value at zero is forced; at one there are two choices.

### MM8036-U7.09 · C · 30 min
Learning objective: U7.2 Semantic substitution

Explain why interpreting $t[\sigma]$ agrees with evaluating the interpretation of $t$ in the environment determined by $\sigma$. Prove the variable and application cases.

**Hint 1 — a starting point.** Substitution and evaluation both proceed structurally.

**Hint 2 — the key move.** For a variable, the tuple component is selected on either side.

### MM8036-U7.10 · A · 50 min
Learning objective: U7.2 Soundness is not completeness

The set model interprets every identity proof between fixed endpoints, when one exists, by the same element. Explain why soundness alone does not prove syntactic UIP.

**Hint 1 — a starting point.** Soundness sends derivable equalities to semantic equalities.

**Hint 2 — the key move.** Its implication points in only one direction.

### MM8036-U7.11 · A · 50 min
Learning objective: U7.1 Explicit substitution design

Why does the equation $(t[\sigma])[\tau] = t[\sigma \circ \tau]$ not by itself prove that an explicit-substitution rewrite system terminates? Construct a simple warning using orientations.

**Hint 1 — a starting point.** An equality can be oriented in either direction.

**Hint 2 — the key move.** Allow both directions and a cycle appears.

### MM8036-U7.12 · A · 50 min
Learning objective: U7.2 Reindexing a family

Let $A = \{0, 1\}$ with fibers as in U7.07, and let $\sigma : \{r, s, t\} \to A$ send $r, s$ to zero and $t$ to one. Describe the reindexed family, its total space, and its number of sections.

**Hint 1 — a starting point.** The fiber at an input is the old fiber at its image.

**Hint 2 — the key move.** Two singleton fibers and one two-element fiber remain.

### MM8036-U7.13 · A · 60 min · Optional challenge

Learning objective: U7 Transfer: When local choices descend

Let $\sigma : \Delta \to \Gamma$ be a surjection of finite sets and let B be a family over Gamma. Characterize exactly which sections s of the reindexed family arise as $s = t \circ \sigma$ for a section t of B. Prove uniqueness and give an example of a section that does not arise this way.

**Hint 1 — a starting point.** Two addresses with the same image must receive the same value.

**Hint 2 — the key move.** Use surjectivity for existence at each old address, and constancy on its fiber for independence of the chosen preimage.

# 8 Unit 8 · Categorical models and proof assistants

**MM8036-U8.01 · E · 15 min**
Learning objective: U8.1 Context morphisms
Explain how contexts and substitutions form a category, naming identities and composition.
**Hint 1 — a starting point.** Use the equations from Unit 7.
**Hint 2 — the key move.** Objects are well-formed contexts; arrows supply their variables.

**MM8036-U8.02 · E · 15 min**
Learning objective: U8.1 Terminal context
Why is the empty context terminal in the substitution category?
**Hint 1 — a starting point.** Count the declarations that need values.
**Hint 2 — the key move.** There is exactly one empty tuple from any context.

**MM8036-U8.03 · E · 15 min**
Learning objective: U8.1 Evaluation
In sets, write evaluation $B^A \times A \to B$ and the curry of $h : X \times A \to B$.
**Hint 1 — a starting point.** An exponential element is a function.
**Hint 2 — the key move.** Fix the first coordinate of $h$.

**MM8036-U8.04 · E · 15 min**
Learning objective: U8.2 Checker contract
What does successful checking of a lab's theorem establish, and what does it leave to the author?
**Hint 1 — a starting point.** Separate the encoded type from the informal intention.
**Hint 2 — the key move.** The checker validates the term against the actual type under its rules.

**MM8036-U8.05 · C · 30 min**
Learning objective: U8.1 Exponential uniqueness
In sets, prove the curry in U8.03 is the unique $k : X \to B^A$ satisfying $\mathrm{eval} \circ (k \times 1_A) = h$.
**Hint 1 — a starting point.** Evaluate a candidate at every $x, a$.
**Hint 2 — the key move.** The equation forces $k(x)(a) = h(x, a)$.

**MM8036-U8.06 · C · 30 min**
Learning objective: U8.1 Beta soundness
Explain categorically why interpreting an application of an abstraction satisfies the beta equation.
**Hint 1 — a starting point.** An abstraction is interpreted by currying.
**Hint 2 — the key move.** Use the evaluation–curry identity and reindex along the argument.

### MM8036-U8.07 · C · 30 min
Learning objective: U8.1 Comprehension in sets

For a set $\Gamma$ and family $A(\gamma)$, construct the context extension and prove its pairing property.
**Hint 1 — a starting point.** Use the total space of the family.
**Hint 2 — the key move.** A point consists of an environment and a value in its fiber.

### MM8036-U8.08 · C · 30 min
Learning objective: U8.2 Debugging an induction

An attempted proof of right-zero addition uses reflexivity for the variable case. Explain the failure and repair it using the lab definitions.
**Hint 1 — a starting point.** The first argument of addition is not a constructor.
**Hint 2 — the key move.** Split into zero and successor and use congruence.

### MM8036-U8.09 · C · 30 min
Learning objective: U8.2 Transport in code

Describe how to cast a vector of length $m$ to one of length $n$ given $p : \mathrm{Id}_{\mathrm{Nat}(m,n)}$. State its reflexive computation.
**Hint 1 — a starting point.** Use the family of vectors at varying lengths.
**Hint 2 — the key move.** Transport in $k \mapsto \mathrm{Vec}(A, k)$.

### MM8036-U8.10 · A · 50 min
Learning objective: U8.1 CCC and dependent models

Explain why a cartesian closed category by itself supplies a model of simple arrows but does not automatically provide all the dependent-type structure used here.
**Hint 1 — a starting point.** Simple arrows use one fixed object for each type.
**Hint 2 — the key move.** Dependent types vary over contexts and must reindex.

### MM8036-U8.11 · A · 50 min
Learning objective: U8.2 Constructive proof audit

Construct a program of type $(A + (A \to 0)) \to (((A \to 0) \to 0) \to A)$. Explain both branches, why this does not establish unrestricted double-negation elimination, and why replacing the missing decision input by a postulate would violate the completed-lab contract.
**Hint 1 — a starting point.** Inspect the supplied decision before using the double negation.
**Hint 2 — the key move.** In the negative branch, the double negation produces an impossible input to empty-type elimination.

### MM8036-U8.12 · A · 50 min
Learning objective: U8.1 Synthesis of substitution and semantics

In the families-of-sets model, prove that reindexing a term along $\sigma : \Delta \to \Gamma$ and then $\tau : \Theta \to \Delta$ equals reindexing along $\sigma \circ \tau$. Relate this to the syntactic explicit-substitution equation.
**Hint 1 — a starting point.** A term is a section $t(\gamma) \in A(\gamma)$.
**Hint 2 — the key move.** Evaluate both constructions at an arbitrary $\theta$.

### MM8036-U8.13 · A · 60 min · Optional challenge

Learning objective: U8 Transfer: A model that forgets a distinction

Interpret the base type o of the simply typed core as a singleton set and c as its element. Give two closed terms of type $o \to o$ with equal interpretations but which are not beta convertible. Explain what this shows about using one model to check equations.

**Hint 1 — a starting point.** Compare the identity function with a constant function.

**Hint 2 — the key move.** Normal forms can be syntactically distinct even when a chosen set has only one possible function to itself.

# 9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

**MM8036-M1.01 · E · 15 min**
Learning objective: U1–U2 Binding and reduction
Compute $\mathrm{FV}((\lambda x.\lambda y.xy)z)$ and reduce the term to normal form.
**Hint 1 — a starting point.** Compute free variables before contraction.
**Hint 2 — the key move.** The free $z$ does not conflict with $y$.

**MM8036-M1.02 · E · 15 min**
Learning objective: U3–U4 Typing and logic
Derive the type $(A \to B) \to A \to B$ and interpret the term as a logical proof.
**Hint 1 — a starting point.** Introduce a function and its input.
**Hint 2 — the key move.** Apply the function to the input.

**MM8036-M1.03 · C · 30 min**
Learning objective: U2–U4 Normalization
Give a weakly but not strongly normalizing term, then explain why it cannot be typable in the arrow-only core.
**Hint 1 — a starting point.** Use an ignored divergent argument.
**Hint 2 — the key move.** Combine the explicit infinite path with strong normalization.

**MM8036-M1.04 · C · 30 min**
Learning objective: U5–U6 Dependent data
Define vector append on paper and justify its result index in both cases.
**Hint 1 — a starting point.** Match on the first vector.
**Hint 2 — the key move.** Its length recursion follows addition on the first argument.

**MM8036-M1.05 · A · 40 min**
Learning objective: U3–U4 Metatheory synthesis
In the arrow-only language with sole constant $c : o$, show every closed term of type $o \to o$, applied to $c$, reduces to $c$. Explain what this does and does not say about equality of functions.
**Hint 1 — a starting point.** Type the application and use base-type canonicity.
**Hint 2 — the key move.** Testing one closed argument is not a general extensionality rule.

**MM8036-M1.06 · A · 40 min**
Learning objective: U5–U8 Proof and model
Construct the dependent-pair map of U5.06 and explain how the same formula acts on total spaces in the set model. Prove it respects the projection to the new base.
**Hint 1 — a starting point.** The first coordinate determines the target fiber.
**Hint 2 — the key move.** Project after mapping and compare with applying $f$ after projection.

# 10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

### MM8036-M2.01 · E · 15 min
Learning objective: U1–U3 Substitution

Reduce $(\lambda x.\lambda z.xz)y$ and give its type under $y : A \to B$.
**Hint 1 — a starting point.** No binder conflicts with the argument.
**Hint 2 — the key move.** The resulting abstraction applies $y$.

### MM8036-M2.02 · E · 15 min
Learning objective: U5–U6 Identity

Give the type and defining reflexive case of transport in a family $P : A \to U_0$.
**Hint 1 — a starting point.** Specify both endpoints.
**Hint 2 — the key move.** At reflexivity it is the identity function.

### MM8036-M2.03 · C · 30 min
Learning objective: U4–U5 Constructive evidence

Construct a term taking $\Sigma_{x:A}(B(x) \times C(x))$ to $(\Sigma_{x:A}B(x)) \times (\Sigma_{x:A}C(x))$. Explain why the reverse direction is generally unavailable.
**Hint 1 — a starting point.** Use the same witness in both output pairs.
**Hint 2 — the key move.** Two independent existential witnesses need not coincide.

### MM8036-M2.04 · C · 30 min
Learning objective: U6 Induction

Prove that mapping the identity function over a vector returns the original vector, using propositional equality.
**Hint 1 — a starting point.** Induct on the vector.
**Hint 2 — the key move.** Use congruence of the fixed-head cons constructor.

### MM8036-M2.05 · A · 40 min
Learning objective: U2–U4 Confluence and typing

Suppose a closed simply typed term reduces along two paths to $u, v$. Prove $u, v$ have a common normal form of the original type.
**Hint 1 — a starting point.** Use preservation, normalization, and confluence in different roles.
**Hint 2 — the key move.** Normalize both reducts and then compare them.

### MM8036-M2.06 · A · 40 min
Learning objective: U6–U8 Equality audit

Explain why the following inference is invalid: "The set model identifies all proofs of equality; therefore the paper theory may use K, and the labs may omit `--without-K`." Give a correct relationship among syntax, semantics, and the lab options.
**Hint 1 — a starting point.** Soundness is one-way, and checker options define the accepted fragment.
**Hint 2 — the key move.** A model can validate additional principles that the syntax does not assume.